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ConGAT: Context-aware graph attention network for 3D region of interest discovery in multiplexed microscopy images[☆]

Hossein Fathollahian^{a,*,*}, Siyuan Zhao^a, Nafiul Nipu^a, Marziye Salahshour^b, Lei Duan^b, Wei Tang^a, Saeed Boorboor^a, G. Elisabeta Marai^a

^a University of Illinois Chicago, Chicago, IL, USA

^b Rush University Medical Center, Chicago, IL, USA

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ABSTRACT

We present ConGAT, CONtext-aware Graph ATtention Network, a visual analysis system for identifying and analyzing cell–cell interactions in large tissue volumes, which leverages graph attention networks and self-supervised learning. Advanced multichannel microscopy techniques, such as Cyclic Immunofluorescence (CycIF), enable detailed, biomarker-based single-cell mapping, but yield large, densely packed volumes that are difficult to explore. ConGAT implements a machine learning-assisted visual analysis approach for these data. First, ConGAT constructs 3D spatial graphs from volumetric image data and applies a 3D graph attention network to model cell–cell relationships directly in 3D space. Next, we introduce a self-supervised feature refinement strategy that learns discriminative region-of-interest representations without labeled data. Through an interactive visual interface, ConGAT allows experts to define arbitrary combinations of biomarkers, enabling flexible ROIs identification and iterative exploration of spatial patterns. We assess ConGAT on synthetic ground truth and on real-world data. An evaluation with domain expert validation demonstrates that ConGAT identifies spatial patterns and candidate ROIs that show substantial agreement with expert assessments of biological relevance.

1. Introduction

A biomarker is an objective, measurable characteristic, such as a protein, gene, or tissue signal, that indicates a biological process, disease state, or response to treatment [1–3]. In biological imaging, biomarkers are particularly useful because their spatial organization can reveal how tumor, immune, stromal, and stress-response processes are arranged within a microenvironment. Cyclic immunofluorescence (CycIF) is a multiplex microscopy technique that measures multiple proteins as biomarkers in a single specimen. It uses repeated cycles of staining, imaging, and signal removal to generate spatially aligned channels, with each channel showing the intensity of a specific protein marker [4,5]. These data support spatially resolved analysis of protein-expression patterns and tissue microenvironments [6–8]. The resulting data are difficult to analyze manually. A single specimen may contain tens of biomarker channels, dense cellular structures, anisotropic sampling, and terabyte-scale volume sizes. Domain experts, therefore, do not simply segment one object class. Rather, they iteratively select sets of biomarkers, search for local neighborhoods in which

the selected signals are co-enriched or spatially adjacent, inspect the raw channels, and decide whether a candidate region is biologically plausible.

Numerous computational approaches have been developed for large-scale multiplexed microscopy analysis, including conventional image analysis, classical machine learning, deep learning, and neural network-based models for characterizing biomarker distributions and interactions across image volumes [9–14]. Yet important limitations remain. 2D methods do not capture the full 3D organization of tissue or the joint behavior of multiple biomarkers across volumetric neighborhoods [15–18]. Recent 3D graph-based approaches, including graph models [19–21], can represent volumetric structure but typically require supervised training with large annotated datasets. Moreover, existing 3D graph neural network applications in biomedical imaging have largely focused on low-channel modalities, such as multidimensional computed tomography (CT) or Magnetic Resonance Imaging (MRI), limiting their suitability for significantly higher-dimensional multiplexed microscopy (in order of tens of biomarker channels).

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* Correspondence to: Computer Design Research and Learning Center (CDRLC) at UIC, 850 W Taylor St, Chicago, IL 60607, United States of America.

E-mail addresses: hfatho2@uic.edu (H. Fathollahian), szhao69@uic.edu (S. Zhao), mnipu2@uic.edu (N. Nipu), Marziye.Salahshour@rush.edu (M. Salahshour), lei_duan@rush.edu (L. Duan), tangw@uic.edu (W. Tang), boorboor@uic.edu (S. Boorboor), gmarai@uic.edu (G.E. Marai).

We introduce ConGAT, a context-aware graph attention framework for 3D ROI discovery in multiplexed microscopy. ConGAT represents a user-selected biomarker context as a graph over composite voxels. Edges encode local 3D proximity, while node features encode marker intensity and the active microenvironment category. This representation supports analytical evidence that is local, spatial, and context-dependent. By assigning learnable attention weights to neighboring composite voxels and biomarker channels, ConGAT captures higher-order 3D biomarker interactions that are difficult to represent with conventional 2D methods. Moreover, graph attention enables learning nonuniform neighbor weights, allowing the model to emphasize informative adjacent regions and de-emphasize weak or noisy neighbors, rather than treating every voxel in a fixed 3D kernel equally.

ConGAT is trained without manual ROI labels. Unlike existing 3D extensions of graph attention models, ConGAT employs a self-supervised training objective that combines contrastive consistency between augmented graph views, masked biomarker reconstruction, and spatial-distance regularization. These objectives encourage embeddings that are robust to missing or weak biomarker signals while preserving 3D tissue geometry. At inference, ConGAT computes category-aware saliency and ROI scores that integrate learned node context, selected-marker intensity, and physical proximity. A high-scoring region indicates co-localization, co-enrichment, or spatial proximity among selected biomarkers, signaling biomarker interaction. Together, these components produce stable node embeddings and category-specific saliency scores for ROI discovery without labeled training datasets, which are often scarce, error-prone, and labor-intensive to collect.

To make the computationally derived ROIs interpretable and actionable for domain experts, ConGAT links graph-based region discovery with interactive 3D visualization. Detected ROIs are overlaid on the original 3D image data, allowing users to inspect their spatial extent, local tissue context, and associated biomarker signatures. The interface supports channel selection, intensity adjustment, opacity control, and interactive spatial navigation, enabling experts to compare marker distributions within and around candidate ROIs. In summary, ConGAT provides an annotation-free visual analytics framework for discovering biologically meaningful ROIs in high-dimensional 3D multiplexed CycIF data. Our main contributions are:

- a self-supervised graph attention approach for identifying protein co-localization and spatial interactions across arbitrary biomarker sets
- a visual computing implementation for detecting subtle and early-stage protein interactions that are often obscured in low-intensity signals or deeper tissue regions
- an interactive 3D analytical interface designed with domain experts, for in situ investigation of spatial biomarker relationships, and
- an expert evaluation with cancer and cell biology researchers demonstrating the method's utility for domain-specific analytical tasks

2. Background and related work

Microscopy regions of interest (ROIs) are biologically enriched tumor, immune, or stromal regions analyzed through multiplex spatial imaging to characterize tissue-level organization and cellular interactions [22]. Existing ROI discovery methods are largely designed for 2D imaging and typically analyze local neighborhoods within individual tissue sections [15,23]. However, many cellular structures and biomarker interactions extend across multiple depths [24,25], limiting the ability of slice-based approaches to capture the full volumetric organization of tissue and potentially overlooking complex 3D immune–tumor microenvironments. As ROI discovery often relies on spatial biomarker patterns, co-localization and protein interactions are central to tissue analysis. Co-localization describes the spatial overlap

of proteins, suggesting potential relationships, whereas protein interaction refers to their physical or functional association [26,27]. Although co-localization alone does not imply interaction and typically requires further validation [28].

Deep neural networks, particularly convolutional neural networks (CNNs), have been widely applied to microscopy and volumetric image analysis. However, CNNs operate on fixed grid structures and do not explicitly model spatial relationships between distant tissue regions. In contrast, graph neural networks (GNNs) represent tissue as connected regions, enabling more natural modeling of complex 3D spatial dependencies [29,30]. The graph attention mechanism further learns the relative importance of neighboring regions, allowing the model to prioritize biologically meaningful interactions within large, highly multiplexed tissue datasets. This adaptive weighting is critical for identifying informative ROIs without manual annotations. Most neural network-based approaches typically require large, well-annotated datasets, motivating the use of self-supervised learning strategies that reduce reliance on manual annotations [31–34]. Furthermore, existing ROI discovery workflows often depend on manual region identification and expert inspection, limiting scalability and introducing observer-dependent variability [35]. These challenges motivate the development of automated methods that provide a standardized framework for prioritizing biologically relevant ROIs and supporting more consistent and repeatable analysis workflows [36].

Graph attention networks (GATs), use learnable attention to weight neighbor contributions, thereby enhancing node representations that capture heterogeneous, context-dependent interactions [29,37,38]. Researchers have applied several attention-based models to biological problems, including protein function prediction, where graph convolutions integrate spatial and biochemical relationships within a radius [39,40]. GAT-based models have been widely applied in spatial omics to identify spatial domains and cell niches, including STGATE, SpaGCN, and SPACE-GM [13,41–44]. Although extensions such as multi-level or co-attention mechanisms improve the modeling of cell–cell interactions [45,46], most of these approaches are largely limited to 2D data or predefined graphs. Several recent approaches have explored graph-based learning in 3D neuroimaging settings, including 3DViT-GAT for sMRI analysis [19], and graph neural network–transformer frameworks for diffusion MRI tractography that integrate local anatomical connectivity with global brain connectivity information, also SC-to-FC studies show that complex models can match group-average baselines [47,48], PointGAT for 3D point-cloud modeling [49], and DGAT-net for capturing local and global dependencies through dynamic graph attention [20]. Additional examples include PointGAT for modeling point clouds via graph attention in 3D object detection [49] and DGAT-net for capturing local and global dependencies in point clouds using dynamic graphs and attention mechanisms [20]. However, most existing methods are supervised graph-learning approaches requiring large annotated datasets. In contrast, ConGAT leverages an attention-based GNN for self-supervised ROI discovery. Moreover, while prior work focuses on MRI-derived anatomical or connectivity graphs, ConGAT is designed for highly multiplexed biological imaging data with high-dimensional biomarker profiles and scarce annotations [50,51].

Self-supervised learning (SSL) is well-suited for graph neural networks in spatial modeling, as the scarcity and cost of annotations [51–53] limit fully supervised approaches. By deriving supervisory signals directly from data, SSL enables effective representation learning without explicit labels and has been successfully applied in biomedical imaging and biological domains [31,54,55]. Recent studies have combined SSL with graph neural networks for ROI discovery in histopathology, typically focusing on 2D data, predefined graphs, or a single learning objective [34,56,57]. Similar, unsupervised representation learning approaches have leveraged autoencoder-based normative modeling of diffusion MRI tract profiles to detect individualized white-matter abnormalities across neurodevelopmental, epilepsy, and schizophrenia

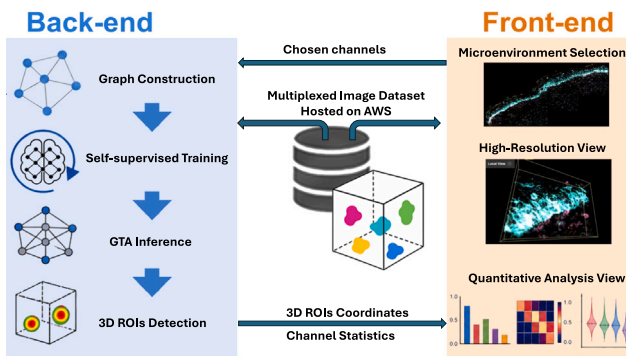


Fig. 1. End-to-end ConGAT architecture linking multiplexed microscopy data processing, graph construction, self-supervised learning, and ROI detection with interactive visual analysis through AWS-based data storage.

cohorts [58]. In contrast, ConGAT leverages SSL learning to refine representations within a single highly multiplexed imaging dataset, enabling the identification of biomarker-specific spatial interactions and informative ROIs.

Co-localization analysis methods fall into intensity-based, overlap-based, and object-based categories, exploiting signal correlation, spatial coincidence, and relationships among segmented entities [59–61]. As biomarker counts grow, these interactions form complex set relationships that are difficult to interpret with traditional spatial methods [62,63]. In contrast, ConGAT integrates multiple biologically relevant criteria for ROI discovery, including spatial proximity, biomarker co-localization, and channel overlap patterns, while leveraging self-supervised learning to enhance the representation of complex cellular interactions.

Multi-volume and visual analysis approaches address the growing need for scalable, interactive exploration of large-scale biological data [64,65], including microscopy volumes [66]. Multi-volume methods improve scalability through adaptive mesh refinement, independent indexing, compressed and neural representations, and hierarchical formats such as OME-Zarr [67–69], but neither identifies ROIs. Visual analysis frameworks such as Napari, Viv, Loom, Vitessece, and Voreen instead support interactive exploration [70–75], but are often weakly integrated with computational model outputs, limiting biological validation and predominantly optimized for 2D workflows with limited 3D interactivity. ConGAT integrates interactive visualization and exploration tools to facilitate efficient identification of biomarker-specific ROIs in large-scale multiplexed 3D tissue images.

3. Methods

Pathology and cancer-biology experts typically reason about ROIs by jointly considering marker co-expression, signal intensity, and spatial proximity [5,18,76]. ConGAT operationalizes this workflow through a self-supervised graph model coupled with a visual analytics system, illustrated in Fig. 1. A 3D spatial graph (Section 3.3) is encoded by a graph attention model (Section 3.4), refined through self-supervised training (Section 3.5), and used to score and extract candidate ROIs (Section 3.6). These candidates are then visualized within the tissue volume through an interactive interface (Section 3.7), enabling experts to explore, compare, and validate the detected regions.

3.1. Design process and domain requirements

ConGAT was developed in collaboration with an interdisciplinary team of domain experts from **medical center**, including clinicians and researchers with expertise in oncology, cancer biology, and cell biology, together with visual computing researchers. The team was initially

formed in response to the Bio + Med + Vis Challenge at IEEE VIS 2025, which called for computational and visualization approaches to support domain experts in exploring large-scale multiplex microscopy data. Following the competition feedback and recognition received, the team continued to iteratively consult on data analysis challenges, discuss biological questions, review intermediate prototypes, and refine both computational and visualization requirements. We followed an Activity-Centered Design (ACD) process [77], an analytical workflow proven effective in scientific, interdisciplinary contexts.

The ACD process identified two high-level scientific activities. The first activity is tissue-microenvironment analysis: experts need to understand how selected protein biomarkers are spatially organized across tumor, immune, and stromal regions. The second activity is local ROI validation: after candidate regions are found, experts need to inspect the raw signal, compare biomarker distributions, and decide whether a candidate region is biologically plausible. It is important to note here that our expert collaborators were particularly familiar with the melanoma-in-situ (MIS) dataset (Section 3.2), which we later used in this study as part of our system evaluation. These activities led to the following task and design requirements.

R1: Support biomarker-driven microenvironment specification

Experts begin their analysis by highlighting biological processes or cell populations of interest, such as T-cell organization, B-cell infiltration, inflammatory activity, oxidative stress, or tumor-associated regions. Marker choices can also be revised during exploration, either by starting from known microenvironment categories or by adding and removing individual biomarkers to test alternative interpretations. Thus, the analysis workflow must support both predefined biologically meaningful marker categories and flexible user-defined biomarker combinations. It must allow experts to revise the active marker set without changing the underlying data model or restarting the full visual-analysis workflow.

R2: Identify biomarker interactions in local 3D neighborhoods

During discussions, it was revealed that biomarker interactions are not only confined to molecular contact but also occur in local tissue regions where selected marker-positive cells, tissue structures, or functional signals were spatially adjacent, co-enriched, overlapping, or organized in a way that suggested a tumor-immune, stromal, inflammatory, or stress-response microenvironment. Therefore, the computational workflow must represent this operational notion of interaction by considering biomarker intensity, co-localization, and physical proximity within local 3D neighborhoods.

R3: Support validation, comparison, and iterative refinement

Following ROI identification, experts further inspect the raw biomarker channels by adjusting intensity ranges and compare candidate regions quantitatively to verify biological plausibility. Moreover, they also needed to record candidate locations and biomarker signatures so that promising regions could be revisited, compared across categories, or used in later analysis. Therefore, the workflow must make candidate ROIs visually inspectable and comparable. It must support linked access to raw signal, spatial context, marker-specific summaries, co-localization evidence, and region-level quantitative signatures, while allowing experts to iteratively refine marker choices, thresholds, and candidate selections.

These requirements guided the design of ConGAT as an integrated computational and visual-analysis workflow that connects biomarker-driven hypothesis specification, local 3D interaction modeling, and expert-guided validation within a single visual-analysis workflow. An expert first uses the visual interface to select a predefined microenvironment category or a custom set of biomarkers that defines the biological hypothesis under investigation (R1). The system then aggregates the selected multichannel CycIF data into composite voxels, so that the analysis operates on local 3D tissue units rather than isolated native voxels (R2). ConGAT constructs a radius-based spatial graph

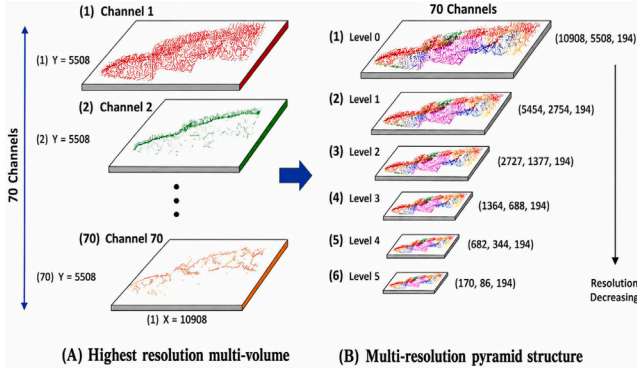


Fig. 2. Melanoma-in-situ dataset structure. (A) multiplexed volume channels. (B) Pyramid resolution, halving x and y at each level.

over these composite voxels, connecting nearby tissue regions that fall within a biologically meaningful local neighborhood (R2). The graph attention encoder computes context-dependent node representations by weighting neighboring composite voxels according to their biomarker signals and spatial relationships. The ROI scoring stage ranks candidate regions by combining category-specific saliency, marker intensity, co-localization, and physical proximity, thereby operationalizing the expert notion of a candidate biomarker interaction region. The expert then reviews the ranked ROIs in the linked 3D interface, inspects the raw biomarker channels, adjusts intensity ranges and visual encodings, compares candidate regions using quantitative summaries, and records promising ROI coordinates and biomarker signatures for further analysis (R3). In this way, ConGAT connects local 3D interaction modeling and ROI volume visualization in a single iterative visual-analysis process.

3.2. Data

This study utilizes a melanoma-in-situ (MIS) dataset comprising patient-derived melanoma tissue imaged in its original spatial context, preserving the arrangement of tumor and immune cells. The data were collected via multiplexed immunofluorescence imaging (CycIF) [5], enabling detailed analysis of the biological microenvironment and immune-tumor interactions within the tissue. The imaging presents the tissue as spatially aligned volumetric channels, each representing a specific biomarker (protein), revealing its spatial distribution and intensity at the single-cell level. The dataset features a large 3D volume measuring $194 \times 5508 \times 10,908$ voxels (3D pixels), with a voxel resolution of approximately $0.1\text{--}0.3 \mu\text{m}$ [78]. It includes approximately 70 biomarker channels totaling approximately 1.6 TB, hosted on AWS S3, with scalable pyramid storage and access at different resolutions (Fig. 2). The multi-resolution pyramid supports scalability and a shared enabling mechanism that lets the same volume serve interactive rendering and graph construction at a tractable scale. The front-end streams the resolution level appropriate to the current zoom, so the 1.6 TB volume can be rendered interactively (R3), while the back-end operates on a downsampled pyramid level whose resolution is matched to the composite-voxel scale, bounding the number of graph nodes and keeping training and inference tractable.

Moreover, these volumes enable the identification and characterization of tissue microenvironments and treatment-related processes [5, 78,79]. Biology-specific microenvironments, such as T-cell maturation, inflammatory zones, regulation of oxidative stress, and B-cell infiltration, are defined based on biomarker expression and spatial proximity. Based on observations and expert knowledge gathered during the ACD workflow, we curated a set of microenvironment categories as ConGAT presets to support category-aware modeling and interpretation

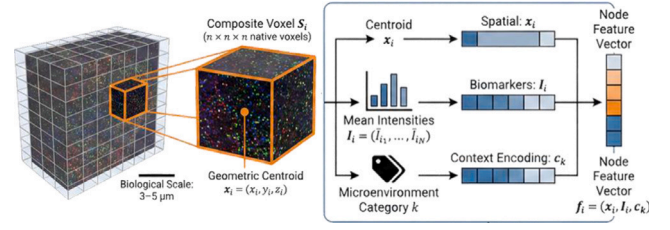


Fig. 3. Composite voxel generation and graph node encoding from spatial, biomarker, and microenvironment features.

of spatial patterns (R1). The associated biomarkers are provided in the Microenvironment Categories Table in the Supplementary Material. Experts can further define and add arbitrary biomarker combinations during exploratory analysis.

3.3. Spatial graph construction

In the clinical workflow, experts identify neighboring tissue regions that jointly express selected biomarkers within a 3D microenvironment, motivating a graph representation where biological evidence is inherently local and spatial. The CycIF input is represented as a tensor $I \in \mathbb{R}^{T \times C \times X \times Y \times Z}$, where T is time, C is the number of biomarker channels, and X, Y, Z are spatial dimensions. Since this study analyzes a single time point, we operate on the multichannel volume $I_i(c, x, y, z)$.

A dense native voxel grid is computationally impractical and too noisy for tissue-scale reasoning. Therefore, the volume is partitioned into non-overlapping composite voxels $S = \{S_1, \dots, S_m\}$, where each S_i is an $n \times n \times n$ block of native voxels with centroid $\mathbf{x}_i = (x_i, y_i, z_i) \in \mathbb{R}^3$. Let $M_k \subseteq \{1, \dots, C\}$ denote the biomarker channels selected for microenvironment category k . Intensities within S_i are averaged after channel-wise normalization, yielding $I_i^{(k)} \in \mathbb{R}^{|M_k|}$. The node feature vector is then:

$$f_i = [\tilde{\mathbf{x}}_i \parallel I_i^{(k)} \parallel c_k], \quad (1)$$

where $\tilde{\mathbf{x}}_i$ is the normalized centroid, c_k is the category encoding, and $\parallel$ denotes concatenation (Fig. 3).

The spatial graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is defined over the composite voxels, where each node $v_i \in \mathcal{V}$ corresponds to S_i with feature f_i and $\mathcal{E}$ represent edge between nodes. Two nodes are connected when their physical distance falls within interaction radius r :

$$\mathcal{E} = \{(v_i, v_j) : i \neq j, d_{ij} \leq r\}, \quad d_{ij} = \|\mathbf{D}(\mathbf{x}_i - \mathbf{x}_j)\|_2 \quad (2)$$

where $\mathbf{D} = \text{diag}(\Delta_x, \Delta_y, \Delta_z)$ accounts for anisotropic voxel spacing. The radius r is chosen to reflect biologically meaningful local tissue neighborhoods on the order of a few μm [5,76], allowing ConGAT to model biomarker co-localization and spatial proximity over 3D tissue units rather than isolated native voxels. For each edge, a bounded inverse-distance coefficient is computed:

$$w_{ij} = \frac{1}{1 + d_{ij}/r} \quad (3)$$

where the +1 term avoids division by zero and stabilizes weighting for interactions within composite voxels that may contain multiple cells. This coefficient is carried forward into the downstream ROI interaction scoring (Eq. (13)).

3.4. Graph attention encoding

Given the spatial graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ constructed from composite voxels, a shared graph attention network (GAT) learns node representations that capture spatial context beyond individual voxels. The attention mechanism adaptively weights neighboring composite voxels

based on feature compatibility, enabling the identification of heterogeneous microenvironmental interactions that are not captured by regular grid-based image analysis. Following the original GAT formulation [29], for attention head h , node features are first linearly projected as

$$\mathbf{h}_i^{(h)} = \mathbf{W}^{(h)} \mathbf{f}_i, \quad (4)$$

where $\mathbf{W}^{(h)}$ is a learnable projection matrix and $\mathbf{f}_i$ is the input feature vector of node v_i (Eq. (1)). For each edge $(v_i, v_j) \in \mathcal{E}$, the attention logit is

$$e_{ij}^{(h)} = \text{LeakyReLU} \left(\mathbf{a}^{(h)\top} \left[\mathbf{h}_i^{(h)} \parallel \mathbf{h}_j^{(h)} \right] \right), \quad (5)$$

where $\mathbf{a}^{(h)}$ is a learnable attention vector and $\parallel$ denotes feature concatenation. The spatial graph structure restricts attention to local 3D tissue neighborhoods, while the inverse-distance coefficient w_{ij} from Eq. (3) is retained for the downstream ROI interaction score rather than used inside the attention logit.

Attention coefficients are obtained via softmax normalization over the spatial neighborhood $\mathcal{N}_i$ of node v_i :

$$\alpha_{ij}^{(h)} = \frac{\exp(e_{ij}^{(h)})}{\sum_{\ell \in \mathcal{N}_i} \exp(e_{i\ell}^{(h)})}. \quad (6)$$

The head-specific embedding of node v_i is obtained by aggregating the transformed features of its neighbors:

$$\mathbf{z}_i^{(h)} = \sigma \left(\sum_{j \in \mathcal{N}_i} \alpha_{ij}^{(h)} \mathbf{h}_j^{(h)} \right), \quad (7)$$

where $\sigma(\cdot)$ denotes the ELU nonlinearity. Outputs from multiple attention heads are concatenated or averaged [29] to yield the final node embedding $\mathbf{z}_i$, representing the local biomarker composition, intensity context, and spatial neighborhood of composite voxel S_i . Architecture details (3 GAT layers, 32 hidden channels, 4 attention heads in the first two layers) are provided in the Supplementary Material.

To obtain interpretable microenvironment-specific relevance scores, ConGAT maps each node embedding to a scalar saliency value via a category-specific projection. For microenvironment category or user-selected biomarker set k , the saliency score is

$$s_i^{(k)} = \sigma(\mathbf{w}_k^\top \mathbf{z}_i), \quad (8)$$

where $\mathbf{w}_k \in \mathbb{R}^{F'}$ is the learnable category-specific projection vector, F' is the embedding dimension, and $\sigma(\cdot)$ here denotes the sigmoid function. The resulting score $s_i^{(k)} \in [0, 1]$ quantifies the relevance of composite voxel S_i to category k , yielding category-aware spatial saliency maps for downstream ROI scoring and region extraction. Additional Implementation details are provided in the Supplementary Material.

3.5. Self-supervised graph encoder training

The GAT encoder is trained to learn node embeddings that are robust to missing or weak biomarker signal while preserving local 3D tissue organization, without manual ROI annotations. A single shared encoder is trained on subgraphs sampled from the neighborhood $\mathcal{N}_i$ of each node v_i , restricted to an interaction radius $r < 5 \mu\text{m}$ [5,76], keeping subgraphs small enough for GPU training while preserving local microenvironment context.

For each sampled subgraph, two stochastic augmented views $\mathcal{G}^{(1)}$ and $\mathcal{G}^{(2)}$ are generated via three perturbations: node-feature masking (removing all biomarker features for a random subset of nodes), biomarker-channel masking (removing selected channels within nodes), and edge dropping (removing a fraction of spatial edges). In our experiments, 10% of node features and biomarker channels are masked, and edges are dropped with probability 0.1. Both views are passed through the same encoder:

$$\mathbf{z}_i^{(1)} = \text{GAT}(\mathcal{G}^{(1)})_i, \quad \mathbf{z}_i^{(2)} = \text{GAT}(\mathcal{G}^{(2)})_i, \quad (9)$$

where $\mathbf{z}_i^{(1)}$ and $\mathbf{z}_i^{(2)}$ are the embeddings of node v_i under the two augmented views. Training jointly minimizes three complementary losses, described below.

Contrastive consistency. encourages the same composite voxel to have similar embeddings across augmented views while distinguishing it from other nodes in the minibatch. For node i and temperature τ ,

$$\mathcal{L}_{\text{contrast}} = -\log \frac{\exp(\text{sim}(\mathbf{z}_i^{(1)}, \mathbf{z}_i^{(2)})/\tau)}{\sum_{\mathbf{z}_j \in \mathcal{B}} \exp(\text{sim}(\mathbf{z}_i^{(1)}, \mathbf{z}_j^{(2)})/\tau)}, \quad (10)$$

where $\text{sim}(\cdot, \cdot)$ is cosine similarity and $\mathcal{B}$ is the set of node embeddings in the sampled subgraph (the training minibatch).

Masked biomarker reconstruction. prevents the encoder from relying on a single strong marker and improves robustness to incomplete biomarker data. A lightweight decoder reconstructs the masked biomarker features $\mathbf{f}_i^{\text{mask}}$ from the node embedding and its neighborhood context, producing reconstruction $\hat{\mathbf{f}}_i^{\text{mask}}$:

$$\mathcal{L}_{\text{mask}} = \left\| \mathbf{f}_i^{\text{mask}} - \hat{\mathbf{f}}_i^{\text{mask}} \right\|^2. \quad (11)$$

Spatial distance regularization. prevents nearby composite voxels from being mapped to arbitrarily distant latent positions, by aligning embedding differences with physical distances over local graph edges:

$$\mathcal{L}_{\text{dist}} = \frac{1}{|\mathcal{E}_B|} \sum_{(i,j) \in \mathcal{E}_B} (\|\mathbf{z}_i - \mathbf{z}_j\|_2 - \alpha d_{ij})^2, \quad (12)$$

where $\mathcal{E}_B$ contains the edges in the minibatch subgraphs, d_{ij} is the anisotropic physical distance (Eq. (2)), and α is a learnable scale relating physical and embedding distances.

Training objective. The GAT encoder is trained by jointly minimizing $\mathcal{L}_{\text{contrast}}$, $\mathcal{L}_{\text{mask}}$, and $\mathcal{L}_{\text{dist}}$. The saliency projection $\mathbf{w}_k$ (Eq. (8)) is trained separately once category-specific supervision becomes available.

3.6. Scoring and region extraction

The trained GAT encoder is applied to the full spatial graph to generate node-level saliency scores $s_i^{(k)}$ for each composite voxel i and microenvironment category k (Eq. (8)). These scores quantify the biological relevance of each voxel to category k but do not, by themselves, capture the strength of physical spatial interactions between neighboring voxels. To identify strong microenvironmental interactions, saliency is combined with biomarker intensity and spatial proximity into a pairwise interaction score.

Pairwise interaction score. For each neighboring pair (v_i, v_j) with $j \in \mathcal{N}_i$, the interaction score for category k combines saliency, biomarker intensity, and spatial proximity:

$$I_{ij}^{(k)} = s_i^{(k)} s_j^{(k)} \bar{t}_i^{(k)} \bar{t}_j^{(k)} w_{ij}, \quad j \in \mathcal{N}_i, \quad (13)$$

where $s_i^{(k)}$ and $s_j^{(k)}$ are the saliency scores of composite voxels i and j for category k ; $\bar{t}_i^{(k)}$ and $\bar{t}_j^{(k)}$ are the corresponding mean biomarker intensities; w_{ij} is the spatial edge weight (Eq. (3)); and $\mathcal{N}_i$ is the set of neighboring composite voxels of node v_i .

Node-level interaction score. Each node is assigned a scalar score by taking the maximum pairwise interaction over its spatial neighborhood:

$$\text{Score}^{(k)}(i) = \max_{j \in \mathcal{N}_i, j \neq i} I_{ij}^{(k)}, \quad (14)$$

representing the strongest biologically meaningful interaction within voxel i 's local neighborhood. Composite voxels are ranked in descending order by $\text{Score}^{(k)}(i)$, and a user-selected threshold (default: top 5%) selects the highest-scoring voxels as candidate high-interaction sites for region extraction.

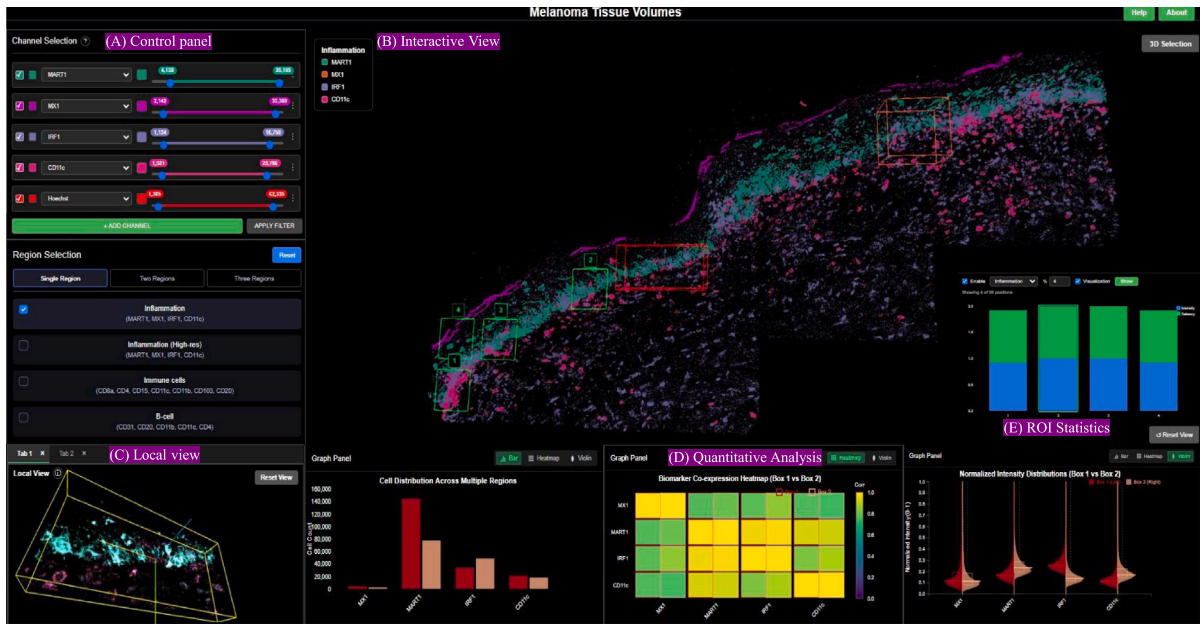


Fig. 4. Interactive visual analytics dashboard for exploring multiplexed 3D microscopy data, detected ROIs, and model explanations. (A) Control panel for biomarker filtering and category selection. (B) Interactive view explores tissue structures and model-detected ROIs in spatial context. (C) Local view inspects selected regions. (D) Quantitative analysis compares voxels, shows biomarker co-expression heatmaps, and regional distributions. (E) ROI statistics rank detections and indicate their saliency.

3.7. ConGAT visualization interface

Fig. 4 presents the ConGAT visualization interface, which follows a coordinated multi-view design supporting biomarker channel exploration, local inspection, quantitative analysis, and model-guided interpretation. Our design leverages scaffolding from visual encodings that are already familiar to the users [80–82]. User interaction and analytical settings are synchronized across all views to maintain a consistent exploration context.

The **control panel** (Fig. 4.A) is the entry point for analysis configuration, where users select biomarkers or expert-predefined categories, adjust intensity thresholds, and specify visualization settings; these configurations propagate throughout the interface and define the context for subsequent analysis. The **interactive view** (Fig. 4.B) provides the primary environment for exploring tissue volumes in 3D, allowing users to navigate, rotate, and zoom the data to examine the spatial organization of biomarker expression while maintaining global context. The **high-resolution view** (Fig. 4.C) supports detailed inspection of selected regions by providing a higher-resolution, localized view of tissue structure in the pyramid dataset, enabling close examination of spatial patterns and biomarker co-localization. The comparison combines the high-resolution view (Fig. 4.C) with the **quantitative analysis module** (Fig. 4.D), which links spatial selections to quantitative summaries, including voxel distributions, co-localization heatmaps, and intensity profiles, to facilitate comparison and characterization of tissue microenvironments. The **ROI analysis view** (Fig. 4.E) presents model-detected coordinates organized into predefined biological categories; selected regions can be overlaid on the interactive view, enabling direct comparison between computational findings and the underlying tissue context to help users investigate and validate model-identified regions.

Since the large number of available biomarkers means predefined categories cannot encompass all biologically meaningful marker combinations, the system also allows users to define custom biomarker combinations and retrieve the corresponding ROI coordinates through the back-end model interface. These results are integrated across all views, supporting flexible exploration of user-defined categories within an analytical workflow.

4. Evaluation and results

When ground truth annotations are difficult or impossible to obtain, it is common to use synthetic data for quantitative evaluation and real-world data for qualitative or indirect evaluation, for example, measuring 2D projection errors for a 3D task [83–86]. Accordingly, we evaluate ConGAT in two complementary ways. First, using a set of synthetic 3D datasets, we evaluate localization accuracy under controlled 3D ground truth, where the true interaction masks are known. Second, using a real clinical dataset, we evaluate whether our model-ranked candidate ROIs are spatially consistent with expert-selected regions and whether the interface supports biological review. Given the lack of real 3D ROI ground truth, we interpret and discuss the results as evidence of expert concordance and plausibility. We additionally provide an ablation study on synthetic 2D data in the Supplementary Materials, examining how individual factors (intensity, proximity, and intersection) affect ROI detection.

4.1. Performance analysis on synthetic ground truth data

Experimental setup. In our experiments, we used an 80/20 train-test split with 100 epochs of self-supervised training. The Adam optimizer used a learning rate of 0.01, contrastive temperature $\tau = 0.1$, and batch size 32. The GAT encoder had three layers with 32 hidden channels, and the first two layers used four attention heads. Local subgraphs were constructed from radius-based spatial neighborhoods. We report performance using IoU (threshold 0.5), F1, mean error (ME), standard deviation (Std), and anisotropic distance (AD). AD is used because the volume is anisotropic, with lower resolution along Z than in the X–Y plane. All models were implemented in Python 3.12 using GPU-accelerated PyTorch on a Windows 11 workstation with an Intel® Core™ Ultra 9 185H processor, 24 GB DDR5 memory, and GPU acceleration for training and inference. Each synthetic configuration took approximately 10 min for both training and evaluation, while the clinical volume took approximately 20 min.

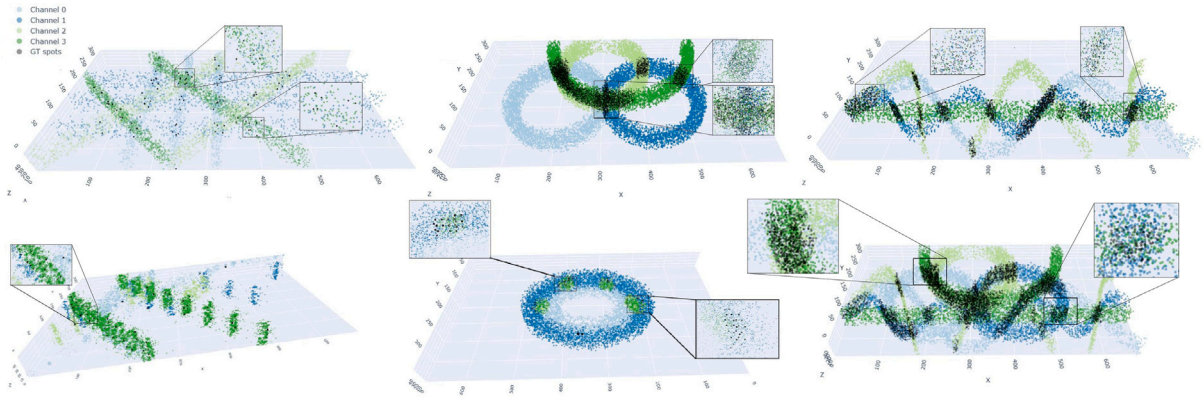


Fig. 5. Synthetic 3D spatial datasets designed to capture proximity-, intensity-, and interaction-based patterns, with colored lines, sinusoids, and circles as interacting features, where the feature points are distributed. Black markers indicate the ground-truth. The last dataset (bottom right) is a composite of the circles and sinusoids datasets, and provides a more complex test case.

Synthetic ground-truth volumes. Since manually annotated 3D multiplexed microscopy ground truth is limited and variable, we constructed synthetic 3D volumes with known interaction masks. The purpose of these volumes was to evaluate whether ConGAT can localize spatial intersections and proximity patterns under controlled geometry, albeit with the limitation that it does not fully reproduce the biological heterogeneity of real tissues. Based on observed structural complexities and variations, validated by domain experts, we generated six texture families that visually resemble the volumetric appearances observed in skin lesions, moles, and similar focal abnormalities, including layered, curved, and intersecting biomarker organizations. Each volume was evaluated under seven non-identity anisotropic scale configurations, yielding 42 volumes.

The six texture families include: (1) linear band structures, comprising four sets of bands with slopes of 0, 1, and -1 , as well as vertical orientations in the x - y plane, producing well-defined intersections; (2) concentric, ring-like bands with similar radii and uniform widths, but differing center locations, resembling Olympic rings; (3) sinusoidal band patterns defined by three frequencies and two amplitudes, including their combinations, with higher frequencies and amplitudes being integer multiples of the lowest; (4) a hybrid pattern that combines three parallel linear-band channels with two oppositely oriented curved channels, overlaid with two additional cloud-like random channels; (5) oval-shaped tubular bands sharing a common center but differing in radius, supplemented by two randomly distributed cloud-like channels with radii comparable to the band widths and randomly varying centers and densities; and (6) a composite Olympic-ring and sinusoidal pattern designed to provide a more complex test case. The textures are illustrated in Fig. 5, respectively.

The synthetic volumes match the composite-voxel scale of the real data, approximately $12 \times 344 \times 681$ composite voxels (Section 3.2), and contain four biomarker-like channels. Intensities are sampled in $[0, 1]$, and channels are generated as band-based structures with fixed width, producing one to four known intersection regions depending on the texture family. To test robustness against anisotropic geometric variation, all combinations of scaling factors 1, 2 are applied independently along the x -, y -, and z -axes, with at least one axis scaled by a factor of 2.

Performance evaluation. Predictions and ground truth are represented as 3D voxel masks at the composite-voxel scale. IoU measures mask overlap, and the F1 score balances precision and recall for interaction-rich region detection. Fig. 6 reports F1 and IoU across texture families, scale configurations, and composite-voxel resolutions. Across the tested textures, ConGAT performs consistently at $8 \times 8 \times 8$ and $16 \times 16 \times 16$ resolutions. The results also show acceptable performance on both stylized textures and more complex hybrid texture combinations, indicating robustness across patterns with different levels of structural

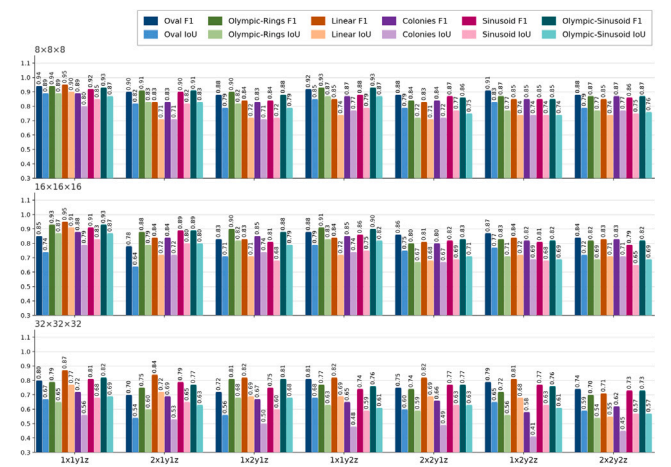


Fig. 6. F1-score and IoU evaluated across synthetic texture patterns, varying anisotropic scaling factors along the x -, y -, and z -axes, and different composite voxel resolutions.

regularity. The $8 \times 8 \times 8$ setting can preserve slightly finer structure but increases the number of graph nodes and computational cost. The $32 \times 32 \times 32$ setting underperforms because over-aggregation merges nearby interaction sites and removes fine spatial structure. Overall, $16 \times 16 \times 16$ provides the best accuracy–efficiency trade-off for the present datasets.

4.2. Performance analysis on a real clinical dataset

Nine domain experts participated in the data evaluation, including six computer science researchers from the University of [X] with varying levels of biomedical visualization experience, and three cancer biology researchers from Medical University [Y]. The biological validity was assessed by three cancer biology researchers. While one evaluator was our domain science collaborator, who contributed domain expertise during system development, the remaining two evaluators were independent and not involved in the development of ConGAT.

The evaluation was designed to measure spatial agreement between expert-selected candidate ROIs and model-ranked candidate ROIs. Experts first received a tutorial on interpreting volume using visual cues, then inspected the tissue volume and selected five ROIs they considered interesting for each queried microenvironment. ConGAT detections were generated independently from the same biomarker context. For each microenvironment category, we compared expert-selected ROIs



Fig. 7. 2D projection of 3D ROIs detected by human experts and ConGAT in the Inflammatory zone.

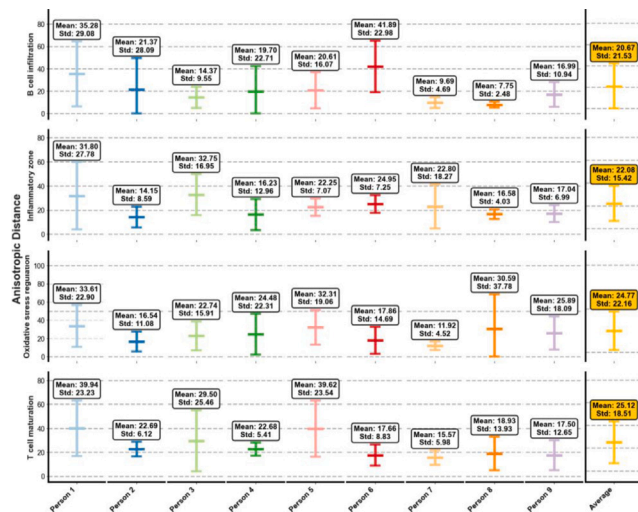


Fig. 8. Agreement between expert-selected ROIs and ConGAT detections across four microenvironment categories (Y axis) for each expert (X axis), measured as the mean and standard deviation of the anisotropic distance (in voxels) from each expert ROI to its nearest ConGAT model detection.

with the five highest-scoring ConGAT ROIs using nearest-neighbor anisotropic distance.

Expert validation was conducted across four predefined microenvironment categories (listed in the Supplementary Material). For each category, experts selected the five most interesting ROIs based on signal intensity, biomarker co-localization, and spatial proximity patterns, and the selected coordinates were compared with the top-ranked coordinates detected by ConGAT. Fig. 7 shows a representative example from the Inflammatory Zone, where colored markers denote expert-selected ROIs and black markers indicate ConGAT detections, demonstrating strong visual agreement between expert selections and model predictions.

To quantify this agreement, Fig. 8 reports the average minimum anisotropic distance from expert-selected ROIs to the nearest ConGAT-detected ROI. The plotted values represent the mean distance, with the corresponding standard deviation computed across all expert-selected coordinates for each microenvironment. Lower distances indicate smaller localization discrepancies and stronger spatial agreement between expert-selected and model-detected ROIs. The consistently low distances across all categories demonstrate that ConGAT identifies correctly those regions consistently selected by human experts. This indicates that ConGAT captures the key spatial patterns used by experts and appropriately prioritizes candidate ROIs for further investigation.

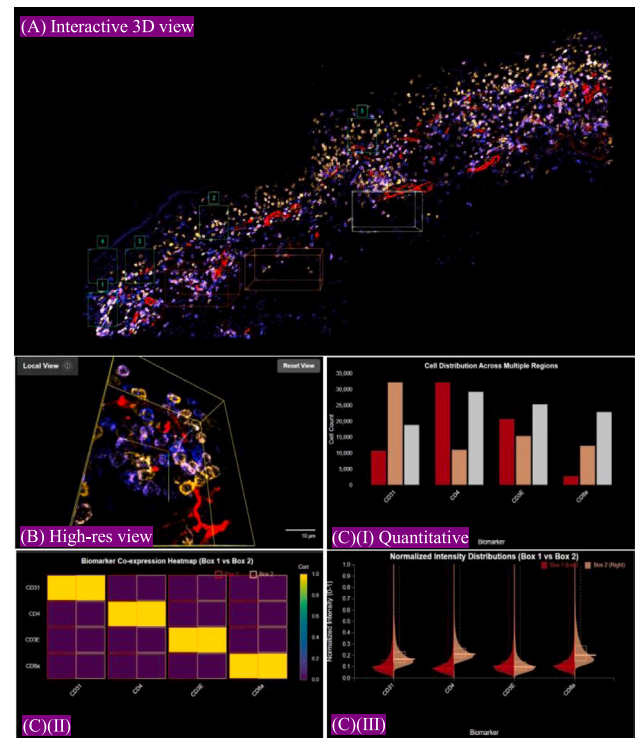


Fig. 9. ConGAT visualization interface for T-cell marker analysis (CD4: blue, CD3E: purple, CD8A: orange, CD31: red). (A) Selected channels with ConGAT ROI overlay and user-defined comparison region. (B) High-resolution view of the selected region for detailed inspection. (C) Quantitative comparison across selected regions: (I) voxel counts per channel, (II) channel co-occurrence, (III) channel intensity distributions.

4.2.1. Case study: Microenvironment investigation

To complement the quantitative evaluation, we conducted two expert-driven case studies to assess how ConGAT supports real-world biological discovery workflows. The first case study examined whether model-prioritized ROIs corresponded to biologically meaningful microenvironments identified through expert analysis, while the second investigated the degree of agreement between expert-selected regions and ConGAT detections when experts explored the data independently of model guidance.

Case study 1: T-cell microenvironment analysis. The expert first activated key T-cell biomarkers, including CD4, CD31, CD3E, and CD8A, and visualized their spatial distribution across the tissue. Next, they enabled the model overlay to inspect the automatically detected T-cell ROIs (Fig. 9(A)). To further examine the model output, the expert selected several candidate regions and compared them using bar charts, co-localization heatmaps, and violin plots, assessing marker expression levels, spatial interaction patterns, and intensity distributions (Fig. 9(B)).

Based on this comparative analysis, the expert selected a representative ROI for detailed inspection. Finally, they explored the selected region using a high-resolution, interactive 3D panel (Fig. 9(C)) to evaluate local spatial organization and marker co-localization. This stepwise analysis identified regions with T-cell clustering and potential cellular interactions, demonstrating the model's ability to detect biologically meaningful microenvironments.

Next, the experts filtered the range and color channels to get more insights, then enabled the ConGAT model overlay and select one of them to detail investigation (Fig. 10(A)), then explore the detected regions by filtering, zooming, and rotating the view (Fig. 10(B), (C), (D)) to check different aspects of interaction. Following their standard

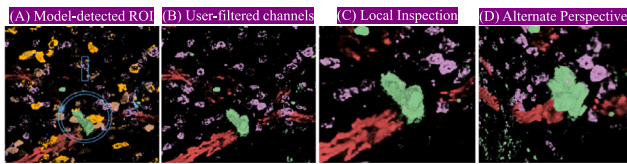


Fig. 10. Case study of expert analysis of a ConGAT-detected T-cell ROI. (A) Detected ROI overlaid on the tissue volume. (B) Visualization of selected biomarkers, including CD8A (green), CD3E (purple), and CD31 (red). (C) Close-up inspection of local spatial organization and co-localization. (D) Rotated 3D view exposing additional spatial relationships within the ROI.

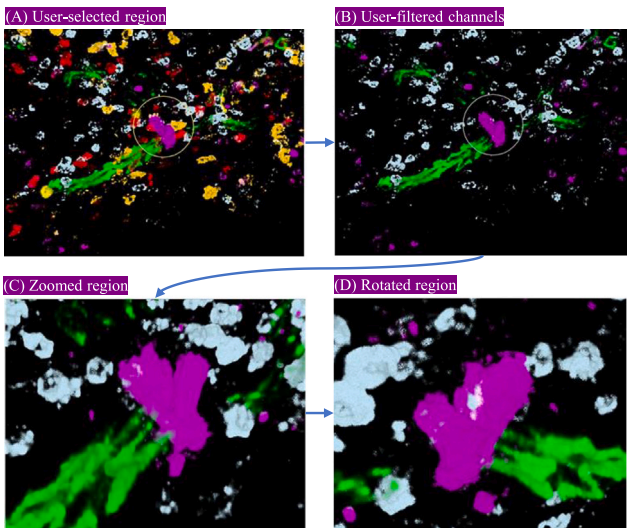


Fig. 11. Case study of a B-cell-rich ROI. (A) User-selected region for analysis. (B) Biomarker channel filtering to reduce occlusion. (C) Zoomed view of local biomarker co-localization. (D) Alternate 3D view revealing additional spatial relationships. Biomarkers are colored as CD19 (cyan), CD20 (purple), CD38 (orange), and Catalase (red).

analysis workflow, the experts confirmed that the highlighted region was biologically meaningful. This agreement with the model-prioritized ROI suggests that ConGAT can effectively identify biologically relevant regions for further investigation.

Case study 2: B-cell microenvironment analysis. In this case study, the expert first explored the tissue volume without model guidance by activating the CD19, CD20, and CD38 biomarker channels and examining their spatial proximity and co-localization patterns (Fig. 11(A, B)). After identifying a candidate B-cell-rich region and investigating it using high-resolution and interactive 3D visualizations (Fig. 11(C,D)), the expert enabled the ConGAT ROI overlay and found that the same region had been detected by the model. The expert subsequently agreed with the model's assessment, demonstrating strong agreement between expert-driven exploration and ConGAT-guided ROI prioritization.

5. Discussion and conclusion

ConGAT addresses the practical problem of finding candidate 3D ROIs in large multiplexed microscopy volumes without requiring voxel-level ROI annotations. The main technical contribution is the integration of a composite-voxel spatial graph network, self-supervised graph attention, and category-aware ROI scoring for arbitrary biomarker contexts. The graph representation preserves local 3D tissue neighborhoods, the attention mechanism learns context-dependent neighbor weights, and the scoring function combines learned saliency with marker intensity and physical proximity. Together, these components

produce ranked candidate regions that experts can inspect, rather than treating the model as an autonomous biological decision-maker.

The evaluation provides complementary evidence. Synthetic experiments show that ConGAT can localize known 2D-3D interaction masks across different texture families. Using real clinical data shows that high-ranked ConGAT ROIs are spatially close to expert-selected candidate ROIs across multiple microenvironment categories. Experts provided strong positive feedback on both the interface and the model-driven ROI detection. They emphasized that the interactive visualization significantly reduced the effort required to identify and verify spatial co-localization patterns, noting that such analysis would be “time-consuming and difficult to perform manually on raw images.” The ability to seamlessly explore biomarker channels, adjust visual parameters, and transition from overview to detailed 3D inspection was highlighted as particularly effective for validating candidate interaction regions. Additionally, the automatic ROI detection was highly valuable, consistently identifying biologically meaningful regions that aligned with expert-driven observations, thereby accelerating analysis and supporting hypothesis generation.

Limitations and future work. Our ConGAT results should be interpreted with appropriate biological caution. ConGAT detects candidate spatial associations among selected biomarker signals and does not establish physical protein interactions, causal mechanisms, clinical relevance, or treatment responses. Co-localization and proximity are useful evidence for hypothesis generation, but downstream biological validation is required. Similarly, the synthetic data provide controlled ground truth for algorithmic localization, but they cannot reproduce the full heterogeneity, staining variability, cellular morphology, and ambiguous boundaries of real tissue microenvironments. While the real clinical data evaluation mitigates this limitation through expert review, including independent cancer-biology evaluators, it remains limited by the number of specimens, the selection of experts, and the number of microenvironment categories.

In terms of the ConGAT technique, the current approach remains computationally intensive, sensitive to sparse marker expression, and dependent on selected hyperparameters, including composite voxel size, graph radius, and ROI threshold. The method also depends on expert-curated biomarker categories or user-selected marker sets, and performance may vary for tissues or marker panels that differ substantially from the melanoma-in-situ data used here. Future work could create larger independently annotated biological datasets for validation, and incorporate weak supervision as partial expert labels become available. Additional work should also improve scalability, support cross-dataset transfer, and extend the interface to guide concrete walkthroughs of use cases in the supplementary material. In conclusion, ConGAT provides an annotation-free framework for candidate 3D ROI discovery in multiplexed microscopy by combining self-supervised graph attention, spatially grounded ROI scoring, and interactive visual validation. By keeping experts in the loop and presenting outputs as candidate spatial biomarker associations, the system supports interpretable exploration of complex tissue microenvironments while avoiding overclaims about biological mechanism.

CRedit authorship contribution statement

Hossein Fathollahian: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization. **Siyuan Zhao:** Writing – review & editing, Writing – original draft, Visualization, Software, Formal analysis. **Nafiu Nip:** Writing – original draft, Visualization, Software, Formal analysis. **Marziye Salahshour:** Validation, Methodology, Conceptualization. **Lei Duan:** Validation, Methodology, Conceptualization. **Wei Tang:** Writing – review & editing, Validation, Methodology. **Saeed Boorboor:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis. **G. Elisabeta Marai:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.cag.2026.104728>.

Data availability

Data will be made available on request.

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